ANGELI WICKRAMA ARACHCHIGE

Data Science Undergraduate | Data Analytics · Machine Learning · Forecasting

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PROFILE

Final-year Data Science undergraduate (SLIIT) with 6 months of industry experience as an AI/ML Engineering Intern at Sri Lanka Telecom, where I built production Python pipelines processing 28.2M daily records across 446,000 customers and Power BI dashboards for operational monitoring. Hands-on across the full analytics lifecycle - data preprocessing and validation, data warehousing and ETL (SQL Server, SSIS, SSAS), Power BI reporting, time-series forecasting (SARIMAX, Prophet), machine learning and anomaly detection, and Generative AI / LLM integration. Currently conducting final-year deep learning research (PyTorch, ESM-2). AWS Certified Cloud Practitioner. Available full-time on weekdays - SLIIT lectures are held on weekends only.

TECHNICAL SKILLS

Programming: Python, SQL, R · Object-Oriented Programming · Data Structures & Algorithms

Data: Structured & unstructured data preprocessing, Data Validation & Quality Checks, ETL/ELT, Data Modeling (Star Schema, SCD Type 2), SQL Server, SSIS, MySQL, MongoDB

BI & Visualization: Power BI, DAX, SSAS OLAP Cubes, Plotly, Matplotlib, Seaborn, Advanced Excel (Power Query, Pivot Tables)

Statistics & Forecasting: Statistical analysis, SARIMAX, Prophet, time-series forecasting

AI / Machine Learning: scikit-learn, Deep Learning (TensorFlow, PyTorch), LightGBM, CatBoost, Anomaly Detection (Isolation Forest, Autoencoders, AWS Random Cut Forest), SHAP explainability, Feature Engineering

Generative AI: LLM integration, prompt engineering.

Backend & APIs: FastAPI, Flask, REST API design & integration, multi-source API aggregation

Cloud & Tools: AWS (S3, Lambda, SageMaker, EventBridge, CloudWatch, SES), Azure fundamentals, Microsoft Fabric (foundational), Docker, Git/GitHub

EDUCATION

BSc (Hons) Information Technology - Data Science Specialization

Sri Lanka Institute of Information Technology (SLIIT) · CGPA 3.40/4.00 · 2023 - Present (Expected : June 2027)

- **Relevant coursework:** Data Warehousing & Business Intelligence (A), Probability & Statistics (A+), Mathematics for Computing (A), Information Retrieval & Web Analytics (A–), Machine Learning & Optimization Methods (A-)

EXPERIENCE

AI/ML Engineering Intern (Lead Intern - FTTH Analytics Team)

Sri Lanka Telecom (SLT) Jan 2026 - July 2026

- Built Python pipelines processing **28.2M daily signal readings across 446,000 customers**, consolidating multi-source operational data (RX/TX signals, ACS exports, fault tickets) into a validated Parquet dataset used by three modelling workstreams.
- Designed automated data-quality checks - invalid-reading removal, per-customer interpolation of **580,195 rows**, and customer-day de-duplication - to guarantee a single trusted source for downstream analytics.
- Developed an **FTTH fault-prediction model** (Isolation Forest, Autoencoders) detecting anomalies **3–7 days before failure**; engineered 55+ temporal features and a 10-rule weighted risk engine, reaching **48% recall at 36% precision** on a 433K-customer holdout month.

- Cut daily false positives from ~378 to **70–90** through 7-day cooldown logic, making the alerts operationally usable by field teams.
- Deployed an end-to-end **AWS pipeline** (S3 → Lambda → SageMaker → EventBridge → CloudWatch → SES) and migrated its feature/preprocessing layer to a new **api_v2 REST schema with zero production disruption**.
- Built **Power BI dashboards** for real-time operational monitoring, documented pipeline and schema changes, and onboarded 3 new team members.

KEY PROJECTS

Global Container Shipment Data Warehouse | *SQL Server, SSIS, SSAS, Power BI, DAX*

- Designed a star-schema data warehouse with SSIS ETL pipelines over 358K+ shipment records from multiple sources (SCD Type 2), deployed as an SSAS OLAP cube with interactive Power BI dashboards.

AirSense - Multi-Country Air Quality Platform | *FastAPI, React, MySQL, SARIMAX, Prophet, LLM*

- Developed a full-stack analytics platform for comparing real-time AQI trends and forecasts across countries and cities, generating time-series forecasts with confidence intervals using SARIMAX and Prophet models.
- Aggregated 4 live APIs with weighted data fusion to produce a unified, comparable AQI signal across regions.
- Integrated LLM-powered narrative reporting over forecast output, auto-generating stakeholder-ready insight summaries with one-click PDF export.

ArtifexAI - Art Auction Price Prediction | *Python, LightGBM, CatBoost, SHAP*

- Developed an end-to-end machine learning solution for art auction price prediction using 60K+ auction records, achieving $R^2 = 0.8449$ and 69.5% of predictions within ±20% error.
- Engineered 50+ textual, image-based, and market-driven features with SHAP explainability for auditable valuations.

Yakadabadu.lk - Scrap Collection & Route Optimization | *MERN Stack, Google Maps API, MongoDB*

- Built the driver and route management module (real-time GPS tracking, Google Maps route optimization) and a scoring algorithm for automated pickup assignment by proximity, load capacity and fuel efficiency.

RESEARCH

AOP-ProCon - Contrastive Learning for Antioxidant Peptide Discovery | *PyTorch, ESM-2*

- Final-year research (in progress): positive-only contrastive representation learning on ESM-2 protein language model embeddings to identify antioxidant peptides - Component 1 of the PlantAOx-RAISE AI framework.
- Curated a tiered training dataset from AODB/AnOxPePred, DFBP and BIOPEP-UWM, and documented the methodology and evaluation design in a formal research proposal and technical analysis reports.

CERTIFICATIONS

- AWS Certified Cloud Practitioner - AWS (2025) · Power BI - Modeling & DAX - Microsoft (2025)
- Data Analysis with R - Google (2025) · AI/ML Engineer Stage 1 & 2 - SLIIT (2025)
- MongoDB Data Modeling Path / Python Developer Path - MongoDB (2026)

LEADERSHIP & ACTIVITIES

- Batch Representative, SLIIT.
- Founder, "Learn DS with Angeli" - free Sinhala-language data science content for Sri Lankan students.
- Volunteer, Rotaract · Volunteer, Women in FOSS · ICT Tutor (A/L & O/L)

LANGUAGES

- Sinhala - Native
- English - Professional Working Proficiency